Daniel Arone • lda@math.ku.dk

Education

University of Copenhagen — Copenhagen, Denmark
PhD Researcher at the Department of Mathematics — *October 2026 - Present*

University of Bonn — Bonn, Germany
Master's Programme in Mathematics — *Winter 2024/2025 - Summer 2026*
MSc in Mathematics
MSc: *Nilpotence in Real Brown–Peterson Theory*

University of Helsinki — Helsinki, Finland
Bachelor's Programme in Mathematical Sciences — *Fall 2021 - Spring 2024*
BSc in Mathematics
BSc thesis: *Roth's Theorem in Diophantine Approximation*

Work and Teaching Experience

TA at the University of Copenhagen
Geometric Topology (NMAK21000U) — *Block 3 of 2026/2027*

Finnish Mathematical Olympiad Training
Lecturing and organization of camps — *2021-2024*
Deputy leader at the Baltic Way Mathematical Contest — *2021*
Deputy leader at the International Mathematical Olympiad — *2023*
Local organizer for the European Mathematical Cup — *2021-2023*

Speaking Experience

1. Nilpotence and Sparseness in the Slice of Real Brown–Peterson Theory.
   Poster presentation at Young Topologists Meeting 2026
2. Graduate Seminar on Global Analysis: K-theory of C^*-algebras. *Fredholm Theory.*
   Winter 2025/2026
3. Graduate Seminar on Topology: Six-Functor Formalisms. *Analytic Manifolds.*
   Winter 2025/2026
4. Graduate Seminar on Advanced Topology: Higher Algebra. *The Lurie Tensor Product.*
   Winter 2025/2026
5. Graduate Seminar on Advanced Geometry: Euler Class and Volume. *Bounded Cohomology.*
   Summer 2025
6. Reading Group on Higher Algebra. $\mathbb{E}_\infty$*-structures* and *Relative Tensor Products.*
   Summer 2025
7. Reading Group on Higher Categories. *The Categories* Pr^L *and* Pr^L_κ.
   Winter 2024/2025

Conference Attendance

European Autumn School in Topology 2026 — *September 14-18, 2026*
New Perspectives on Stable Homotopy and Beyond — *September 7-11, 2026*
Young Topologists Meeting 2026 — *June 29-July 3, 2026*
Kleine AT II: Topological Cyclic Homology — *April 10, 2026*
36. NRW Topologist Meeting — *November 6-7, 2025*
Kleine AT I: Spectral Sequences and the Even Filtration — *October 10, 2025*
Masterclass: Arithmetic and Homotopy Theory — *June 9-13, 2025*